Project Design Phase

Solution Architecture

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Project Name	Strategic Product Placement Analysis: Unveiling Sales Impact with Tableau Visualization

Solution Architecture:

1 Data Sources Layer

- Point of Sale (POS) Systems – Transaction-level sales data (product, quantity, revenue, date, store).
- Inventory Database – Stock levels, product categories, shelf allocation.
- Store Layout Data – Product placement zones (entrance, aisle-end, eye-level shelves, checkout counters).
- Promotional Data – Discounts, campaigns, in-store offers.

2 Data Integration & Preparation Layer

- Data Extraction – Connect Tableau to SQL databases, Excel files, or cloud storage.
- Data Cleaning – Remove duplicates, standardize product names, handle missing values.
- Data Transformation –
- Create calculated fields (Sales Growth %, Average Basket Size, Placement Impact Score).
- Merge sales data with placement zones.
- Data Aggregation – Summarize by product, category, placement zone, and time period.

3 Data Storage Layer

- Centralized Data Warehouse / Data Mart
- Structured tables for:
- Sales Facts
- Product Dimensions

- Store Dimensions
- Placement Dimensions

4️⃣ **Analytics & Business Logic Layer**

- KPI Calculations:
- Total Sales
- Sales by Placement Zone
- Conversion Rate
- Revenue per Square Foot
- Before vs After Placement Comparison
- Trend Analysis (Weekly / Monthly / Seasonal)
- Correlation analysis between placement position and sales performance

5️⃣ **Visualization Layer (Using Tableau)**

Interactive dashboards built in Tableau:

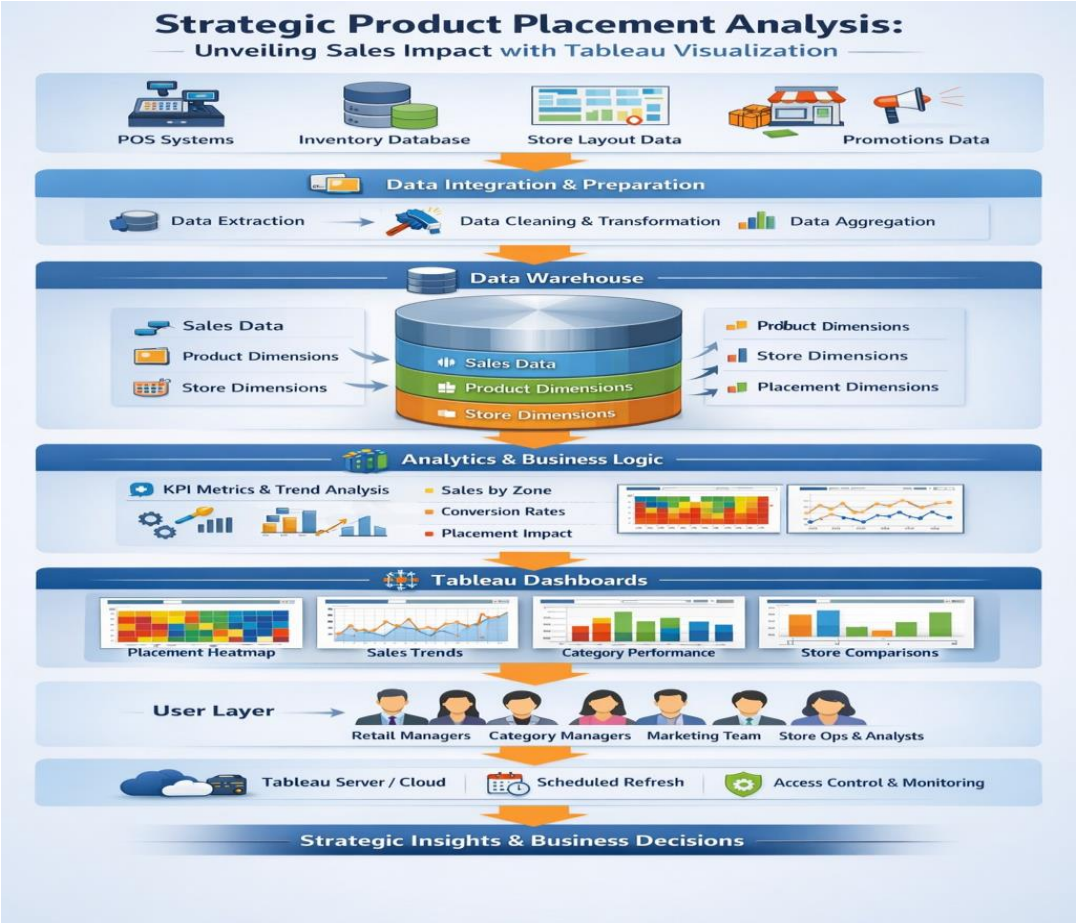
- 📊 Sales by Placement Heatmap
- 📈 Trend Analysis Line Charts
- 🛒 Product Category Performance Dashboard
- 🏬 Store-wise Placement Impact Comparison
- 📍 Shelf Position vs Revenue Bar Charts

6️⃣ User Layer

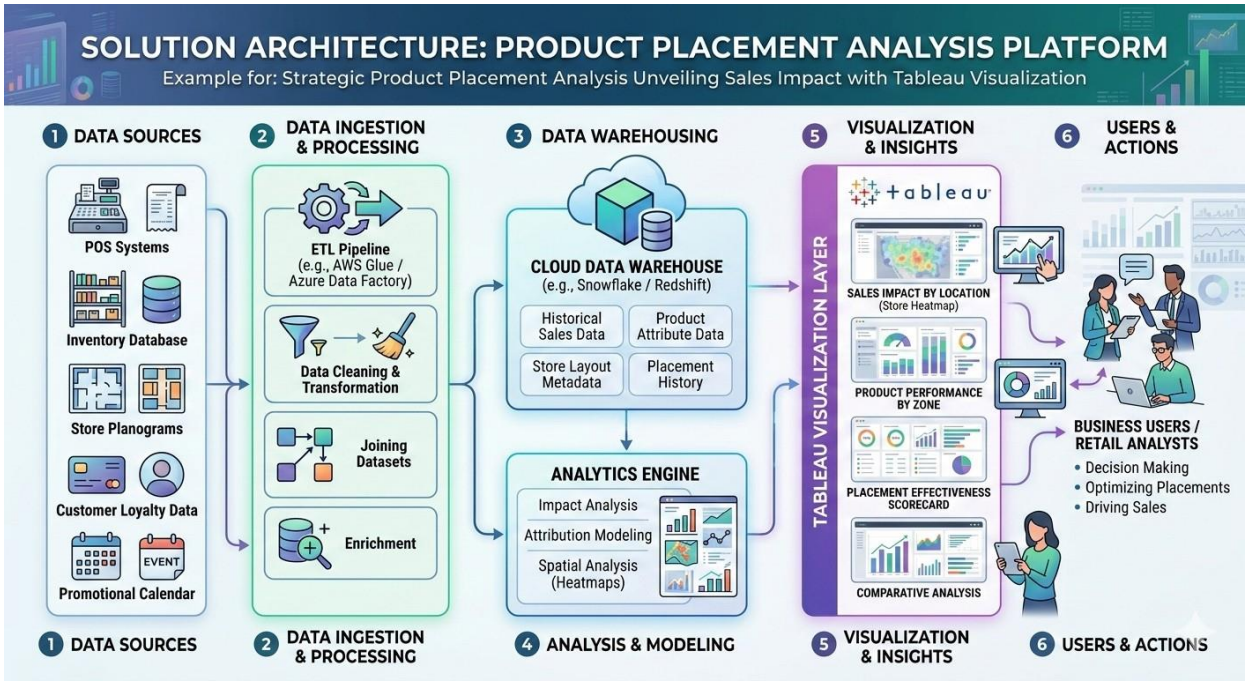
- Retail Managers
- Category Managers
- Marketing Team
- Store Operations Team
- Business Analysts

7️⃣ Deployment & Maintenance

- Publish dashboards to Tableau Server / Tableau Cloud
- Schedule automatic data refresh
- Role-based access control
- Performance monitoring and periodic optimization



Example - Solution Architecture Diagram:


SOLUTION ARCHITECTURE: PRODUCT PLACEMENT ANALYSIS PLATFORM
Example for: Strategic Product Placement Analysis Unveiling Sales Impact with Tableau Visualization
1 DATA SOURCES
2 DATA INGESTION & PROCESSING
3 DATA WAREHOUSING
5 VISUALIZATION & INSIGHTS
6 USERS & ACTIONS
POS Systems
Inventory Database
Store Planograms
Customer Loyalty Data
Promotional Calendar
EVENT
ETL Pipeline
(e.g., AWS Glue / Azure Data Factory)
Data Cleaning & Transformation
Joining Datasets
Enrichment
CLOUD DATA WAREHOUSE
(e.g., Snowflake / Redshift)
Historical Sales Data
Product Attribute Data
Store Layout Metadata
Placement History
ANALYTICS ENGINE
Impact Analysis
Attribution Modeling
Spatial Analysis (Heatmaps)
TABLEAU VISUALIZATION LAYER
+ tableau
SALES IMPACT BY LOCATION
(Store Heatmap)
PRODUCT PERFORMANCE BY ZONE
PLACEMENT EFFECTIVENESS SCORECARD
COMPARATIVE ANALYSIS
BUSINESS USERS / RETAIL ANALYSTS
• Decision Making
• Optimizing Placements
• Driving Sales
1 DATA SOURCES
2 DATA INGESTION & PROCESSING
4 ANALYSIS & MODELING
5 VISUALIZATION & INSIGHTS
6 USERS & ACTIONS